

Supplementary Data for
GERM-BO: a reproducible border-aware adapter framework for genomic foundation
models

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1 Mathematical proofs

This note contains the full mathematical details referenced in the main text. Each proof begins with a one-sentence interpretation.

1.1 Proof of the cluster-length proposition

Interpretation. This proof shows that the border score is a direct measure of expected local burst length.

Proof. Under the renewal cluster model, a cluster continues after each active occurrence with probability ρ_w and stops with probability $1 - \rho_w$. Hence

$$\mathbb{P}(C_w = k) = (1 - \rho_w)\rho_w^{k-1}, \quad k = 1, 2, \dots \quad (1)$$

which is the geometric distribution on $\{1, 2, \dots\}$. Therefore

$$\mathbb{E}[C_w] = \sum_{k=1}^{\infty} k(1 - \rho_w)\rho_w^{k-1} = \frac{1}{1 - \rho_w}. \quad (2)$$

Using the definition $\beta_w = \rho_w/(1 - \rho_w)$ yields

$$\frac{1}{1 - \rho_w} = 1 + \frac{\rho_w}{1 - \rho_w} = 1 + \beta_w. \quad (3)$$

This completes the proof. $\square$

1.2 Proof of the variance proposition

Interpretation. This proof shows that valid self-overlap creates positive local dependence and therefore larger variance of token counts.

Proof. For brevity write $I_t = I_t(w; X)$. Since $N_w = \sum_{t=1}^{T-m+1} I_t$, the variance decomposition gives

$$\text{Var}(N_w) = \sum_{t=1}^{T-m+1} \text{Var}(I_t) + 2 \sum_{1 \leq s < t \leq T-m+1} \text{Cov}(I_s, I_t). \quad (4)$$

Stationarity implies $\mathbb{P}(I_t = 1) = p_w$ and $\text{Var}(I_t) = p_w(1 - p_w)$.

Now fix a shift ℓ with $1 \leq \ell < m$. If $\ell \notin S(w)$, then two occurrences of w at positions t and $t + \ell$ are incompatible because the required prefix and suffix letters disagree. Hence

$$\mathbb{P}(I_t = 1, I_{t+\ell} = 1) = 0, \quad (5)$$

and

$$\text{Cov}(I_t, I_{t+\ell}) = -p_w^2. \quad (6)$$

If $\ell \in S(w)$, then the joint event is feasible and

$$\begin{aligned} \mathbb{P}(I_t = 1, I_{t+\ell} = 1) &= \mathbb{P}(I_t = 1)\mathbb{P}(I_{t+\ell} = 1 \mid I_t = 1) \\ &= p_w q_w(\ell). \end{aligned} \quad (7)$$

Therefore

$$\text{Cov}(I_t, I_{t+\ell}) = p_w q_w(\ell) - p_w^2. \quad (8)$$

Summing over all pairs with the same shift gives

$$\begin{aligned}\text{Var}(N_w) &= (T - m + 1)p_w(1 - p_w) \\ &\quad + 2 \sum_{\ell=1}^{m-1} (T - m + 1 - \ell) \text{Cov}(I_t, I_{t+\ell}) \\ &\quad + R_{\text{long}},\end{aligned}\tag{9}$$

where R_{long} collects shifts at least m . Dropping all nonpositive invalid overlap terms and all nonnegative contributions from R_{long} yields the lower bound used in the main retained-information argument. This completes the proof. $\square$

1.3 Proof of the retained-information theorem

Interpretation. This proof shows that a larger border score lowers retained information because suppression reacts to the larger burst scale of the response.

Proof. By definition,

$$I_w(\tau) = \mathbb{E}[\phi_\tau(a_w)^2 g_w^2].\tag{10}$$

Since g_w is independent of a_w and $\mathbb{E}[g_w^2] < \infty$,

$$I_w(\tau) = \mathbb{E}[g_w^2] \mathbb{E}[\phi_\tau(a_w)^2].\tag{11}$$

Likewise,

$$I_w(\infty) = \mathbb{E}[g_w^2] \mathbb{E}[\phi_\infty(a_w)^2].\tag{12}$$

Because $\phi_\infty(u) = 1$ by definition of no suppression, we obtain

$$I_w(\infty) = \mathbb{E}[g_w^2].\tag{13}$$

Hence

$$R_w(\tau) = \frac{I_w(\tau)}{I_w(\infty)} = \mathbb{E}[\phi_\tau(a_w)^2].\tag{14}$$

Using the scale model $a_w \stackrel{d}{=} b_w Z$ yields

$$R_w(\tau) = \mathbb{E}[\phi_\tau(b_w Z)^2].\tag{15}$$

To prove monotonicity, let $b_1 \leq b_2$. Since $\phi_\tau(u)$ is nonincreasing in $|u|$ and $|b_1 Z| \leq |b_2 Z|$ pointwise whenever $|Z|$ is fixed, we have

$$\phi_\tau(b_1 Z)^2 \geq \phi_\tau(b_2 Z)^2 \quad \text{almost surely.}\tag{16}$$

Taking expectations gives

$$\mathbb{E}[\phi_\tau(b_1 Z)^2] \geq \mathbb{E}[\phi_\tau(b_2 Z)^2].\tag{17}$$

Thus $R_w(\tau)$ is nonincreasing in b_w . Since $b_w = b_0(1 + \beta_w)$ with $b_0 > 0$, $R_w(\tau)$ is also nonincreasing in β_w . If ϕ_τ is strictly decreasing on a set of positive probability, then the inequality is strict for any $b_1 < b_2$ on a set of positive probability. This completes the proof. $\square$

1.4 Proof of the compensation-weight theorem

Interpretation. This proof shows that the compensation weight should grow exactly where retained information becomes small.

Proof. The compensation objective is separable across words. For a fixed word $w \in \mathcal{H}$ define

$$J_w(\gamma_w) = \omega_w \frac{c_w}{\gamma_w^2 R_w(\tau)} + \lambda_\Gamma \gamma_w^2, \quad \gamma_w > 0. \quad (18)$$

Differentiating gives

$$J'_w(\gamma_w) = -2\omega_w \frac{c_w}{\gamma_w^3 R_w(\tau)} + 2\lambda_\Gamma \gamma_w. \quad (19)$$

Setting the derivative to zero yields

$$\lambda_\Gamma \gamma_w^4 = \frac{\omega_w c_w}{R_w(\tau)}. \quad (20)$$

Since $\gamma_w > 0$, the unique stationary point is

$$\gamma_w^* = \left(\frac{\omega_w c_w}{\lambda_\Gamma R_w(\tau)} \right)^{1/4}. \quad (21)$$

The second derivative is

$$J''_w(\gamma_w) = 6\omega_w \frac{c_w}{\gamma_w^4 R_w(\tau)} + 2\lambda_\Gamma > 0, \quad (22)$$

so the stationary point is the unique global minimizer. Finally, because γ_w^* is proportional to $R_w(\tau)^{-1/4}$, any decrease of $R_w(\tau)$ with β_w implies a nondecreasing dependence of γ_w^* on β_w . This completes the proof. $\square$

1.5 Markov continuation mass

For a valid shift $\ell \in S(w)$, under a first-order Markov chain,

$$q_w(\ell) = \prod_{j=m-\ell+1}^{m-1} P_{w_j, w_{j+1}}, \quad (23)$$

up to the shared overlap block fixed by $I_t(w) = 1$. Continuation mass can be estimated by

$$\hat{\rho}_w = \sum_{\ell \in S(w)} \prod_{j=m-\ell+1}^{m-1} \hat{P}_{w_j, w_{j+1}}, \quad (24)$$

where $\hat{P}$ is the empirical transition matrix from a background corpus. On heterogeneous genomes, $\hat{\rho}_w$ should be interpreted cautiously in repetitive or low-complexity regions.

1.6 Implementation mapping from theory to trainer

Tokenizer-to-word alignment. Each tokenizer token $w \in \mathcal{V}$ is mapped to a DNA string. Overlap set $S(w)$ and border score $\beta_w = \hat{\rho}_w / (1 - \hat{\rho}_w)$ (when $\hat{\rho}_w < 1$) are computed on that string.

Controlled metadata score. On synthetic border-aware splits, each sample stores a scalar `border_score` in CSV metadata. During training, the score is batch-normalized and passed to the compensation rule with λ_{comp} , c_{min} , and c_{max} from the YAML config.

External sequence-estimated score. When metadata is unavailable, a label-free center-window k -mer JSD score is computed without class labels, then train-quantile normalized before compensation.

Low-rank branch. GERM-BO replaces $y = Wx + \text{scale} \cdot B(Ax)$ with $y = Wx + \text{scale} \cdot B(Ax \cdot c(x))$, where $c(x)$ is broadcast across valid tokens. No separate Γ matrix is stored.

Hyperparameter correspondence. Config field `compensation_strength` equals λ_{comp} . Fields `compensation_clip_min/max` equal $c_{\text{min}}/c_{\text{max}}$. Theoretical λ_{Γ} , τ , and α are not exposed as trainer knobs in the current release.

2 Additional experimental results

2.1 Mechanism validation

Table 1: Mechanism validation on held-out *border_medium + border_hard* (Supplementary Table S1).

Variant	Accuracy	F1
No compensation	0.8706 ± 0.0685	0.8741 ± 0.0601
Activation-derived compensation	0.8757 ± 0.0543	0.8768 ± 0.0532
Metadata-driven compensation	0.9272 ± 0.0583	0.9251 ± 0.0613
Shuffled metadata	0.8638 ± 0.0598	0.8607 ± 0.0718

2.2 Cross-backbone oracle-metadata replication

Table 2: Cross-backbone replication on oracle-metadata *border_hard* (seeds 50–54; Supplementary Table S2).

Backbone / Method	Seeds	Accuracy	Macro-F1
NT v2 50M LoRA	5	0.8945 ± 0.0124	0.8944 ± 0.0124
NT v2 50M GERM-BO (Meta)	5	0.9828 ± 0.0045	0.9828 ± 0.0045
HyenaDNA tiny LoRA	5	0.9625 ± 0.0090	0.9625 ± 0.0090
HyenaDNA tiny GERM-BO (Meta)	5	0.9945 ± 0.0021	0.9945 ± 0.0021

2.3 External full comparison on the strict splice split

2.4 Class recovery

2.5 Promoter and chromatin null/near-null regimes

On Genomic Benchmarks *human_nontata_promoters* (held-out seeds 45–49), LoRA reaches 0.8178 ± 0.0034 accuracy and label-free GERM-BO reaches 0.8168 ± 0.0032 ($\Delta = -0.0010$).

DNABERT-2 histone and enhancer pilots (pilot seeds 42–44; held-out 45–49) do not establish a robust label-free advantage: no held-out task has a bootstrap CI excluding zero, and pilot/held-out sign agreement is limited. NT v2 50M histone pilots similarly show positive mean deltas on only three of ten marks. Label-free strict-splice transfer is robust on DNABERT-2 but inconclusive on NT v2 50M and near-null on HyenaDNA tiny (main text Discussion).

Table 3: External full comparison on the strict *3-mer-balanced* splice split (Supplementary Table S3).

Method	Accuracy	Macro-F1
3-mer Linear SVM	0.4137 $\pm$ 0.0026	0.4128 $\pm$ 0.0026
3-mer Logistic Regression	0.4094 $\pm$ 0.0023	0.4052 $\pm$ 0.0023
GERM-BO quantile [0.8, 1.2]	0.3888 $\pm$ 0.0233	0.3580 $\pm$ 0.0545
GERM-BO activation-derived	0.3709 $\pm$ 0.0144	0.3112 $\pm$ 0.0492
GERM-BO shuffled metadata	0.3656 $\pm$ 0.0273	0.2844 $\pm$ 0.0842
Gated LoRA	0.3606 $\pm$ 0.0254	0.2769 $\pm$ 0.0873
LoRA attention.output + classifier	0.3489 $\pm$ 0.0081	0.2405 $\pm$ 0.0145
GERM-BO comp=0	0.3489 $\pm$ 0.0081	0.2405 $\pm$ 0.0145
DNABERT-2 frozen linear probe	0.3334 $\pm$ 0.0006	0.1688 $\pm$ 0.0042

Table 4: Per-class performance on the strict splice split, seeds 50–59 (Supplementary Table S4).

Class	F1 score			Prediction count		
	LoRA	GERM-BO	Δ	LoRA	GERM-BO	Δ
Acceptor	0.0842	0.3121	+0.2280	91.5	593.2	+501.7
Donor	0.1684	0.2233	+0.0549	293.1	354.9	+61.8
Non-splice	0.4492	0.4380	-0.0112	1415.4	851.9	-563.5

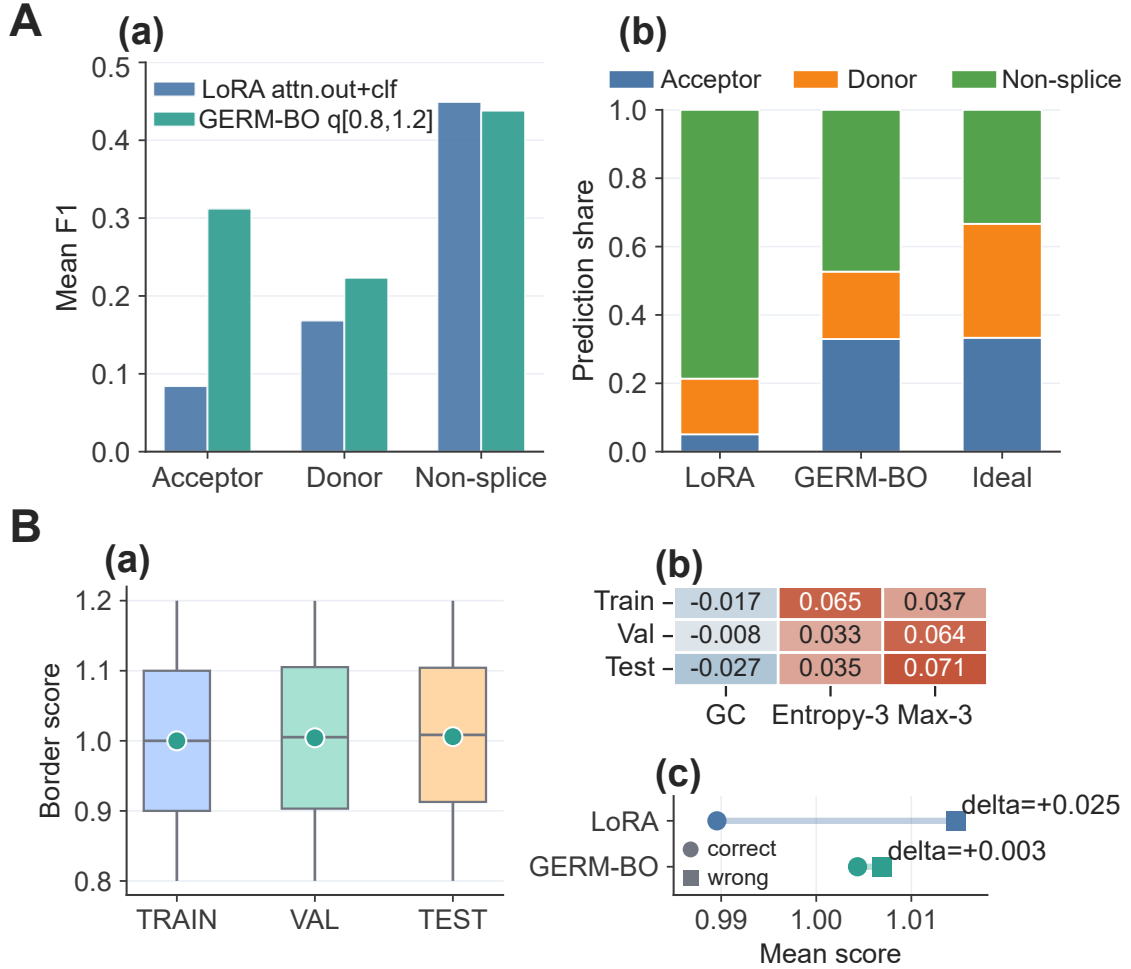


Figure 1: Class recovery and estimator quality on the strict splice split (Supplementary Fig. S2).

A

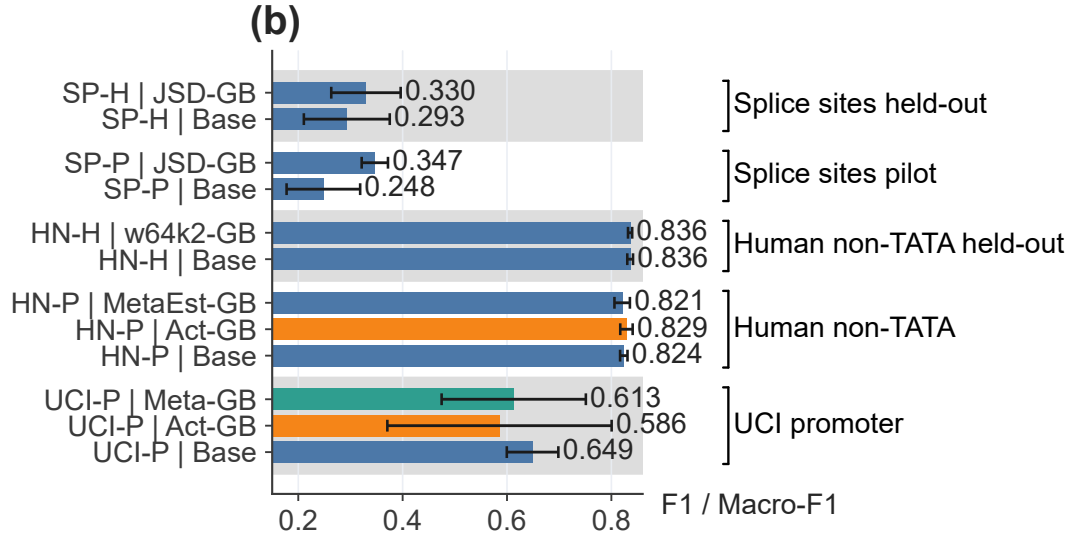
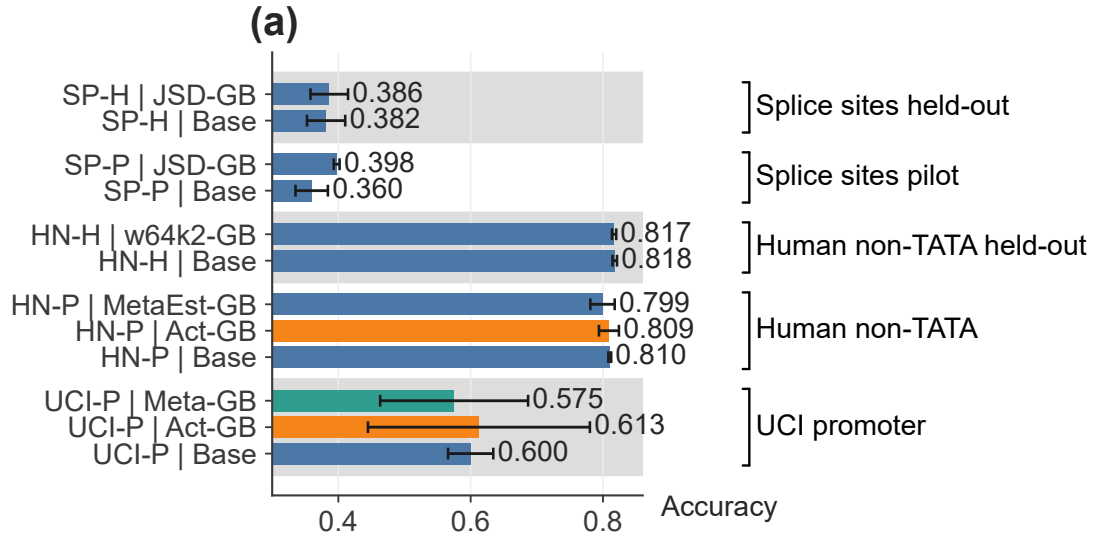


Figure 2: External pilot overview across promoter and splice datasets (Supplementary Fig. S1).

2.6 Sensitivity and resource footprint

On held-out controlled splits, $\lambda_{\text{comp}} \in \{0.15, 0.20, 0.25, 0.27\}$ is stable; $\lambda_{\text{comp}} = 0.27$ is near-optimal on *border_hard*. Rank $r \in \{4, 8, 16\}$ remains competitive at the default $r = 8$. On the strict splice setting (seed 50), GERM-BO and LoRA share the same trainable parameter count (30,744) and peak memory (587 MB); end-to-end times are comparable because compensation is elementwise scaling on the LoRA branch.